USER MANUAL

1. Introduction

Welcome to your **Smart Medication Management System**! This device ensures you never miss a dose by:

✅ **Automated dispensing** at preset times
✅ **Blynk app alerts** (Wi-Fi mode)
✅ **Offline manual control** (No Wi-Fi needed)
✅ **Real-time status tracking** (LCD + LED feedback)

2. Device Overview

Hardware Components

- **ESP32 Microcontroller**: Brain of the system.
- **Servo Motors**: Dispense pills (carousel + gate).
- **IR Motion Sensor**: Detects pill pickup.
- **LCD Screen**: Shows time/dose status.
- **RGB LED**: Color-coded alerts (blue=standby, red=alarm, green=taken).
- **Buzzer**: Audible alarm for dose reminders.

Software Features

- **Blynk IoT App**: Remote control via smartphone.
- **4 Dose Slots**: Schedule up to 4 daily doses. But extedable.
- **Dual Mode**: Works online (Wi-Fi) or offline (buttons).

3. Setup Guide

Step 1: Power On

1. Connect the device to a **5V power supply**.
2. The LCD will display:

"Smart Medication System"

3. The **RGB LED will glow blue** (standby mode).

Step 2: Wi-Fi Connection

- The device auto-connects to Wi-Fi (*installed in the* code).
- On success: LCD shows *"WiFi Connected!"* with the local IP.
- On failure: *"Connection Failed"* (falls back to offline mode).

4. Setting Alarms

Via Blynk App (Online Mode)

1. Open the **Blynk app** and select your project.
2. Use these virtual pins:
    - **V0-V1**: Set alarm hour/minute.
    - **V2**: Toggle AM/PM.
    - **V8-V11**: Select dose slot (1-4).
3. Tap **"Manual Dispense" (V5)** to test the mechanism.

Via Physical Buttons (Offline Mode)

Button	Function
Button 1	Toggle hour/minute setting.
Button 2	Increase time value.
Button 3	Decrease time value.
Button 4	Cycle through dose slots (1-4).
Button 5	Manual dispense (instant dose).

5. Using the System

During an Alarm

1. When a dose is due:
    - **Buzzer sounds** + **LED turns red**.
    - LCD shows: *"Take Medicine Dose X"*.
    - Servos rotate to dispense pills.
2. **After taking pills**:
    - IR sensor detects motion → LED turns green.
    - LCD updates to *"Medicine Taken"*.

Missed Dose Alert

- If pills aren't picked up within **1 minute**:
    - Blynk app notification: *"Missed Dose X"*.
    - LED flashes blue.

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6. Troubleshooting

Issue	**Solution**
No Wi-Fi connection	Check ssid/pass in code. Use offline mode.
Pills not dispensing	Check servo connections (pins 18/19).
LCD not working	Verify I2C address (0x27 in code).
False "Missed Dose"	Clean IR sensor or adjust sensitivity.

7. Safety & Maintenance

- **Power**: Use only **5V adapters**.
- **Cleaning**: Wipe the pill tray with a dry cloth.
- **Storage**: Keep away from moisture/dust.

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8. For More, Scan the QR ↷